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Studierichting: Informatica
Oefeningenreeks 5G nr. 10

Klein teken onderzoek:

$$y = 0 \Rightarrow 0 = \frac{x^2(1+x)}{1-x} \Rightarrow x = 0 \text{ en } x = -1$$

$$f(x) = \frac{x^2 + x^3}{1-x}$$

$$f'(x) = \frac{(2x + 3x^2)(1-x) + (x^2 + x^3)}{(1-x)^2}$$

$$= \frac{2x + 3x^2 - 2x^2 - 3x^3 + x^2 + x^3}{(1-x)^2}$$

$$= \frac{2x + 2x^2 - 2x^3}{(1-x)^2} \Rightarrow x^2 - x - 1 = 0$$

$$\Delta = 1^2 - 4 \cdot 1 \cdot (-1) = 5$$

$$x = \frac{-(-1) \pm \sqrt{5}}{2 \cdot 1} = \frac{1 \pm \sqrt{5}}{2}$$

$$\frac{1 - \sqrt{5}}{2} \approx -0.618$$

$$y^2 = \frac{(-0.618)^2 \cdot (1 + (-0.618))}{1 - (-0.618)} \approx 0.09 \Rightarrow y \approx 0.3$$

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Berekening:

$$V = \pi \int_{-1}^0 \frac{x^2(1+x)}{1-x} dx$$

$$t = 1 - x \Rightarrow dt = -dx \Rightarrow -dt = dx$$

$$\begin{aligned}
&= \pi \int_2^1 \frac{(1-t)^2 \cdot (1+(1-t))}{t} (-dt) \\
&= \pi \int_1^2 \frac{(t^2 - 2t + 1) \cdot (2-t)}{t} dt \\
&= \pi \int_1^2 \frac{(-t^3 + 4t^2 - 5t + 2)}{t} dt \\
&= \pi \int_1^2 \left(-t^2 + 4t - 5 + \frac{2}{t}\right) dt \\
&= \pi \left[-\frac{t^3}{3} + 2t^2 - 5t + 2\ln(t) \right]_1^2 \\
&= \pi \left(-\frac{8}{3} + 8 - 10 + 2\ln(2) + \frac{1}{3} - 2 + 5 + 2\ln|1| \right) \\
&= \pi \left(2\ln|2| - \frac{14}{3} + \frac{10}{3} \right) \\
&= \pi \left(2\ln|2| - \frac{4}{3} \right)
\end{aligned}$$

References